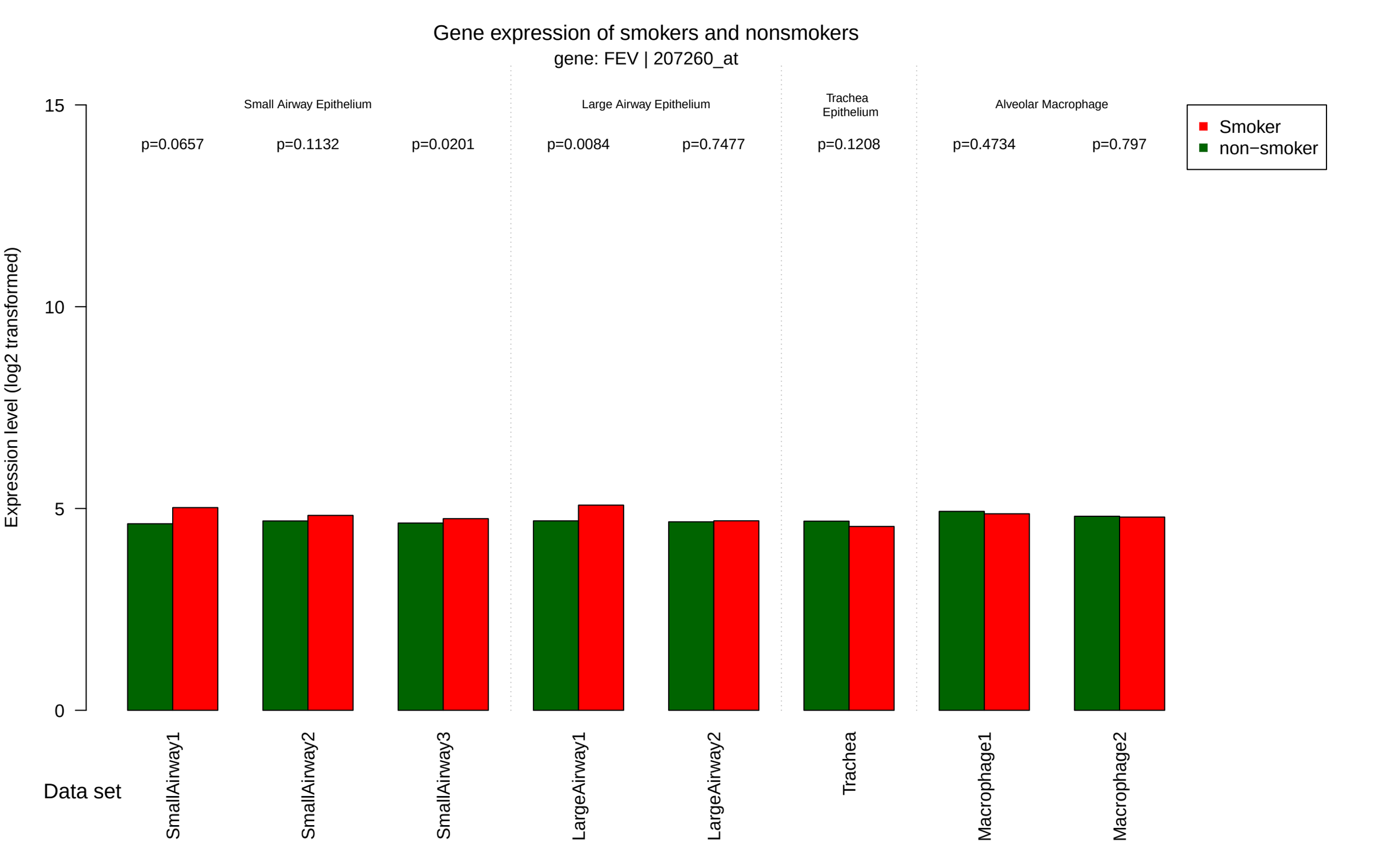
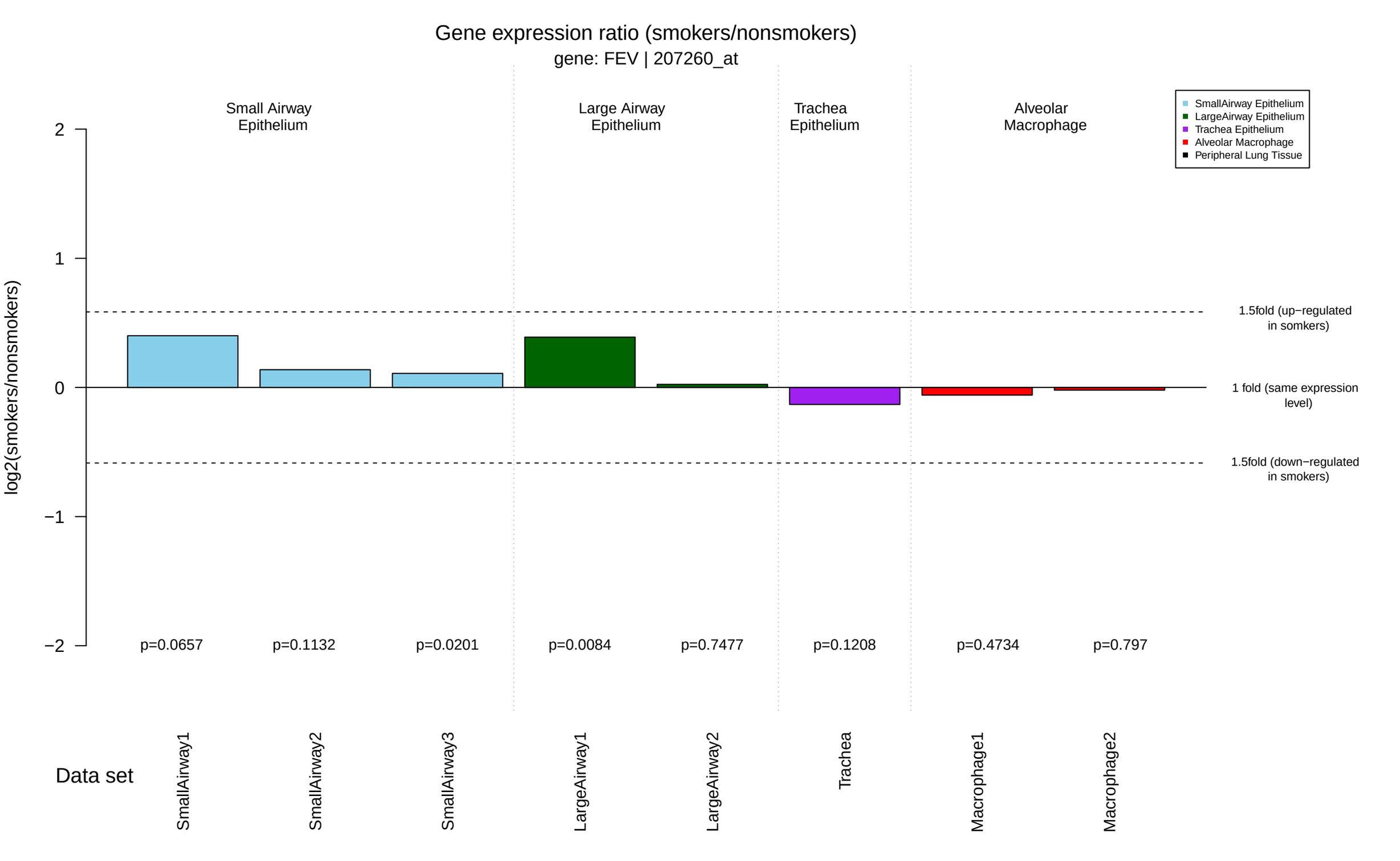
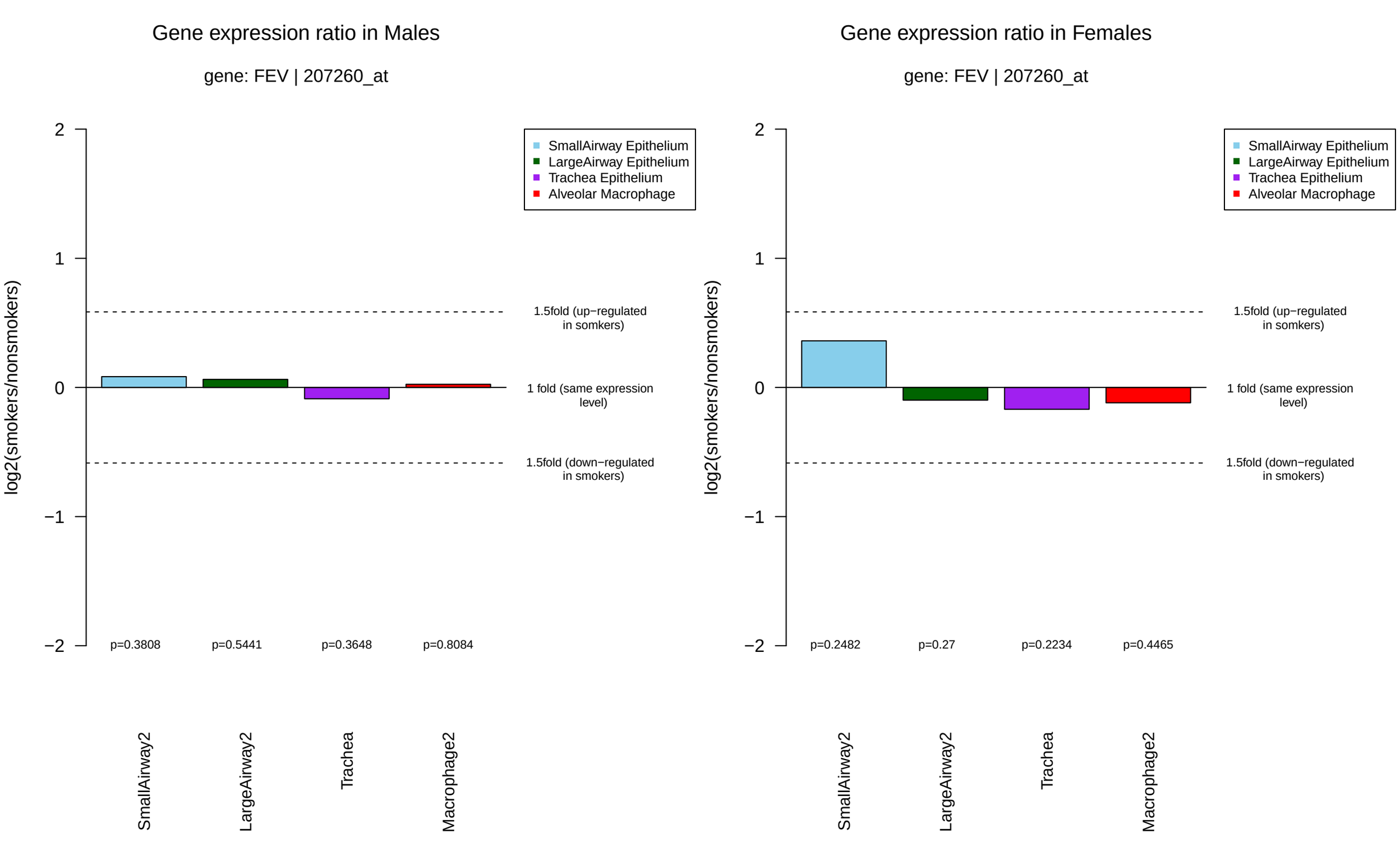
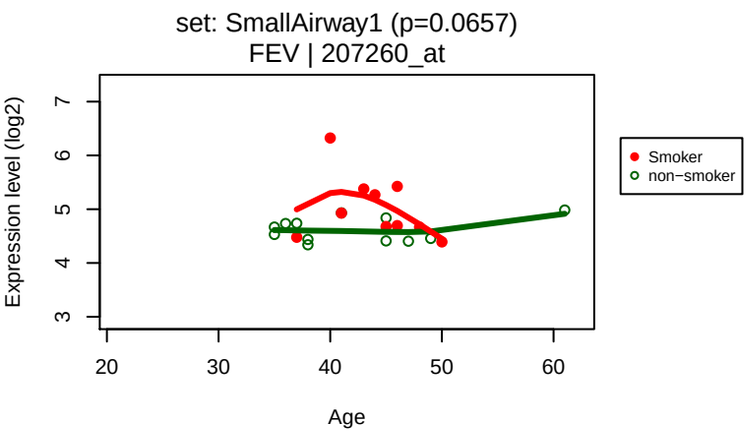
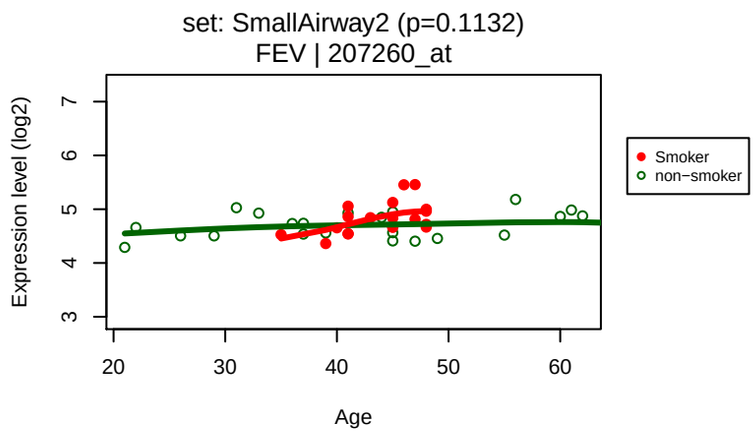
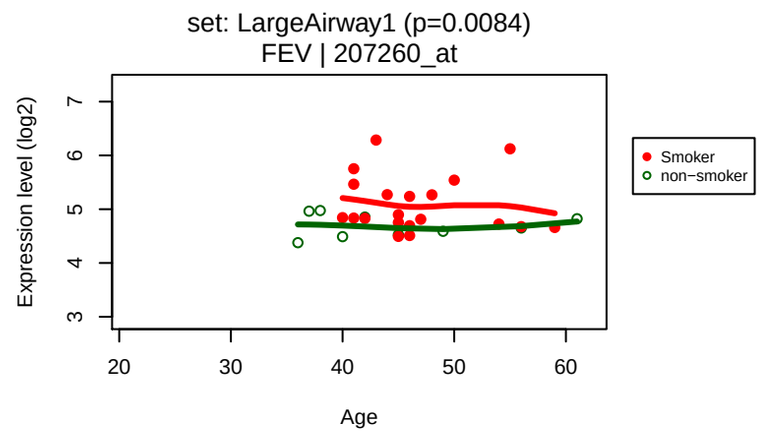

Gene expression of smokers and nonsmokers
gene: FEV | 207260_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0657
p=0.1132
p=0.0201
p=0.0084
p=0.7477
p=0.1208
p=0.4734
p=0.797
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

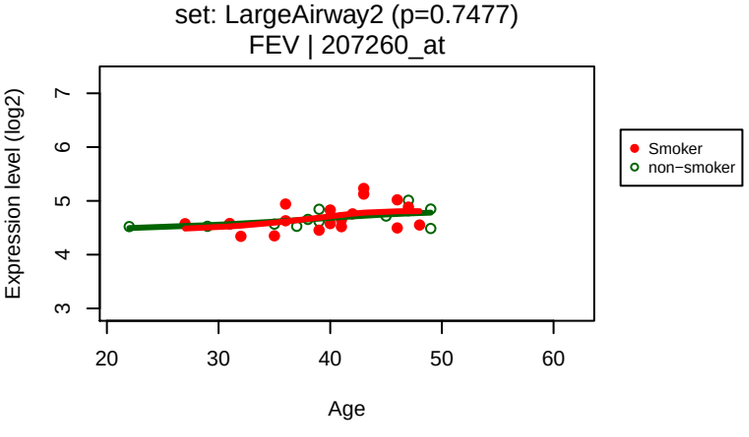

Gene expression ratio (smokers/nonsmokers)
gene: FEV | 207260_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0657
p=0.1132
p=0.0201
p=0.0084
p=0.7477
p=0.1208
p=0.4734
p=0.797
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

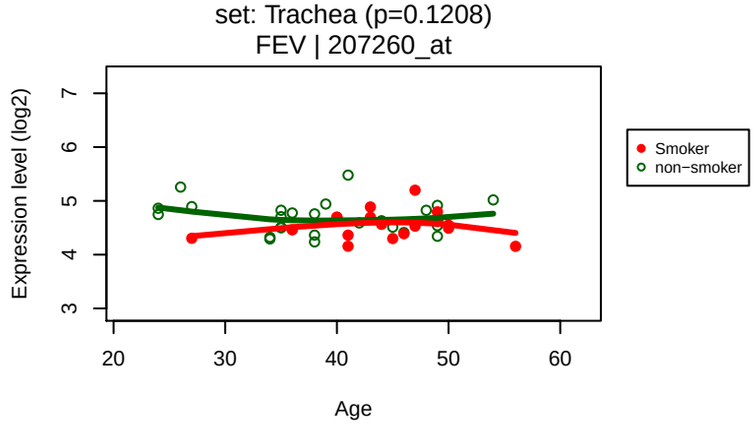

Gene expression ratio in Males
gene: FEV | 207260_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3808
p=0.5441
p=0.3648
p=0.8084
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FEV | 207260_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2482
p=0.27
p=0.2234
p=0.4465
SmallAirway2
LargeAirway2
Trachea
Macrophage2

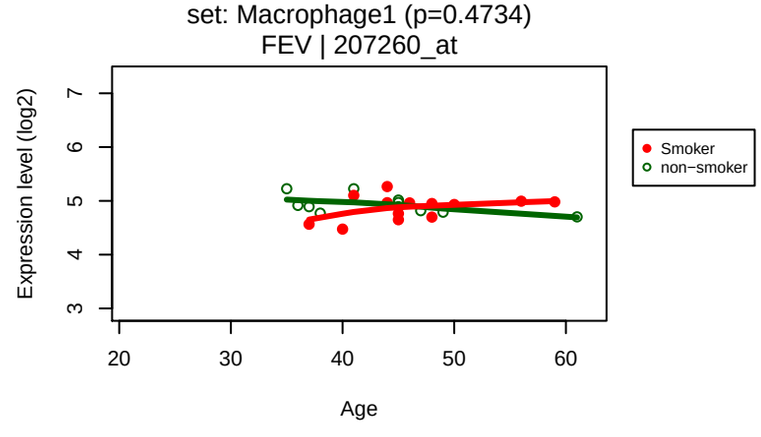

set: SmallAirway1 (p=0.0657)
FEV | 207260_at
Expression level (log2)
Age
Smoker
non−smoker

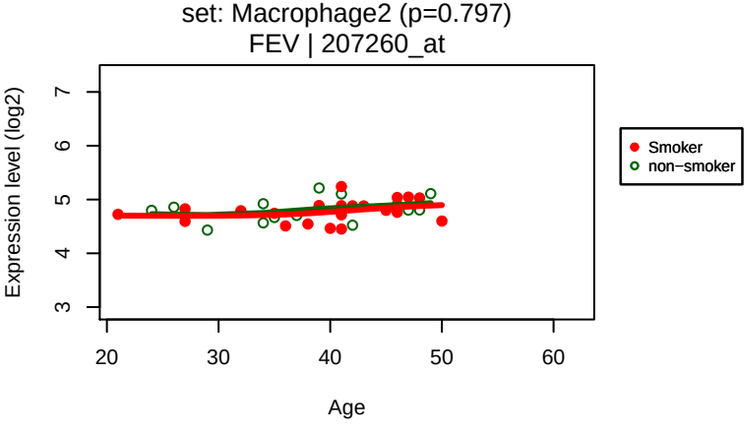

set: SmallAirway2 (p=0.1132)
FEV | 207260_at
Expression level (log2)
Age
Smoker
non−smoker

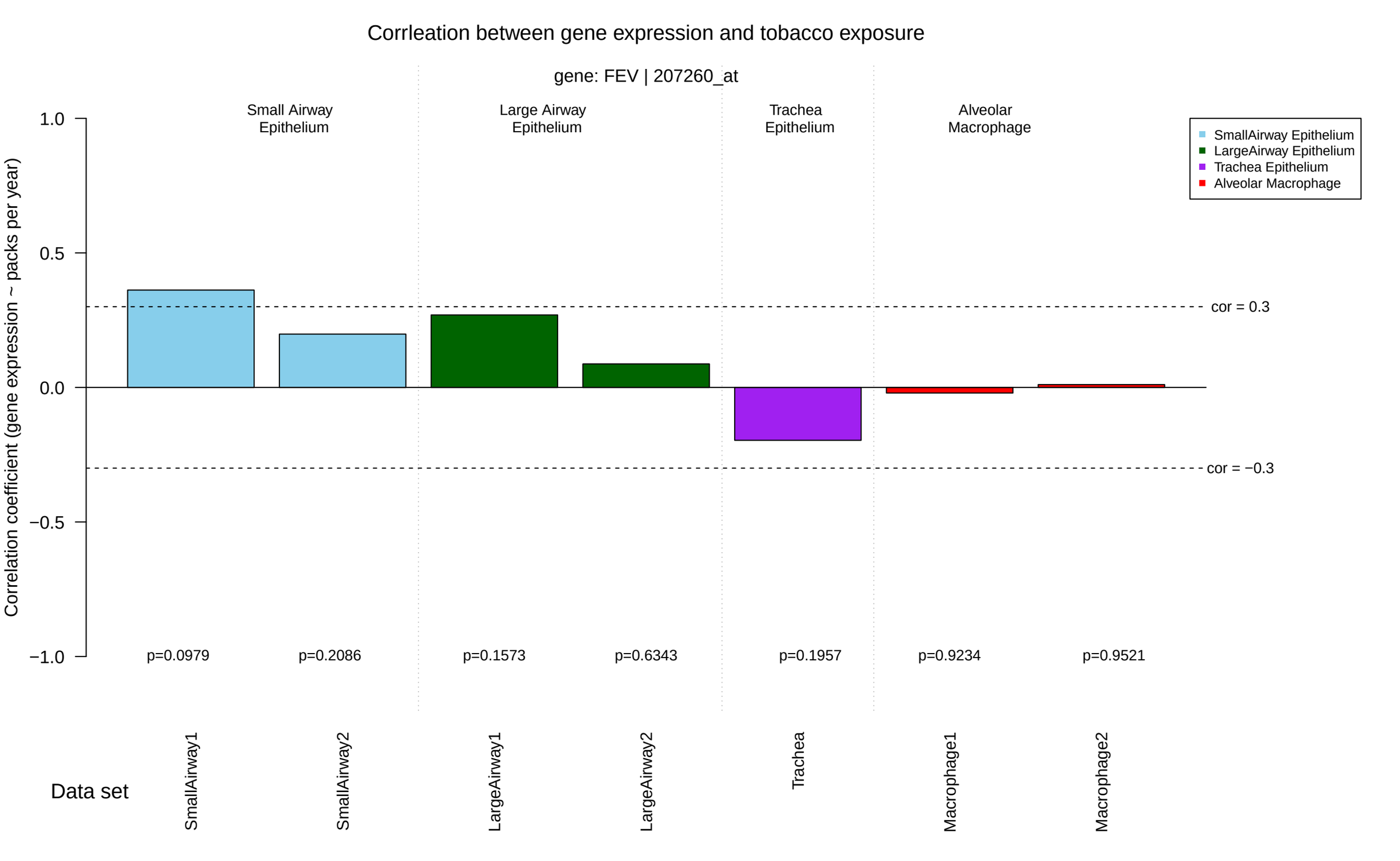

set: LargeAirway1 (p=0.0084)
FEV | 207260_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7477)
FEV | 207260_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1208)
FEV | 207260_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4734)
FEV | 207260_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.797)
FEV | 207260_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FEV | 207260_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0979
p=0.2086
p=0.1573
p=0.6343
p=0.1957
p=0.9234
p=0.9521
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2